

5.1 Warm-Up Exercises

In the following exercises, all coordinates and components are given with respect to a right-oriented orthonormal frame $\mathcal{K} = (O, \mathcal{B})$ where $\mathcal{B} = (\mathbf{i}, \mathbf{j}, \mathbf{k})$.

5.1. Consider the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Calculate $\mathbf{a} \times \mathbf{b}$.

5.2. Determine the distances between opposite sides of a parallelogram spanned by the vectors $\overrightarrow{AB}(6, 0, 1)$ and $\overrightarrow{AC}(1.5, 2, 1)$.

5.3. Consider the vectors $\mathbf{a}(2, -3, 1)$, $\mathbf{b}(-3, 1, 2)$ and $\mathbf{c}(1, 2, 3)$. Calculate $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

5.4. The points $A(1, 2, -1)$, $B(0, 1, 5)$, $C(-1, 2, 1)$ and $D(2, 1, 3)$ are given with respect to an orthonormal coordinate system. Are the four points coplanar?

5.5. Consider the tetrahedron with vertices $A(2, -1, 1)$, $B(5, 5, 4)$, $C(3, 2, -1)$ and $D(4, 1, 3)$. Determine

- a) the volume of the tetrahedron.
- b) the common perpendicular of the sides AB and CD .

5.6. Consider the vectors $\mathbf{a}(2, 3, -1)$ and $\mathbf{b}(1, -1, 3)$. Determine the vector $\mathbf{p}$ which is orthogonal to $\mathbf{a}$ and $\mathbf{b}$ and for which $\langle \mathbf{p}, 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k} \rangle = 51$.

5.7. Consider the points $A(2, 1, -1)$, $B(3, 0, 1)$, $C(2, -1, 3)$ and $D \in Oy$. Determine the coordinates of D if the volume of a tetrahedron $ABCD$ is 5.

5.8. Let π be a plane with normal vector $(1, 1, 1)$ intersecting the coordinate axes in A , B and C . Give formulas for the volume of the tetrahedron $OABC$ in terms of $h = d(O, \pi)$, $a = |OA|$ and $s = |AB|$ respectively.

5.9. Determine the matrices of the linear maps $\phi : \mathbb{V}^3 \rightarrow \mathbb{V}^3$ defined by $\phi(\mathbf{v}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{v}$.

5.10. Consider the vectors $\mathbf{u} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{k}$. Determine the matrix of the linear map $\phi : \mathbb{V}^3 \rightarrow \mathbb{R}$ defined by $\phi(\mathbf{v}) = [\mathbf{v}, \mathbf{u}, \mathbf{w}]$.

5.1. Consider the vectors $\mathbf{a} = \mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$. Calculate $\mathbf{a} \times \mathbf{b}$.

$$\mathbf{a} \times \mathbf{b} = (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}) \times (7\mathbf{i} + 4\mathbf{j} + 6\mathbf{k})$$

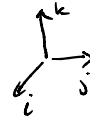
$$= \underbrace{7\mathbf{i} \times \mathbf{i}}_{=0} + 4\mathbf{i} \times \mathbf{j} + 6\mathbf{i} \times \mathbf{k} +$$

$$+ 14\mathbf{j} \times \mathbf{i} + \underbrace{8\mathbf{j} \times \mathbf{j}}_{=0} + 12\mathbf{j} \times \mathbf{k} +$$

$$- 14\mathbf{k} \times \mathbf{i} - 8\mathbf{k} \times \mathbf{j} - \underbrace{12\mathbf{k} \times \mathbf{k}}_{=0}$$

$$= -10 \underbrace{\mathbf{i} \times \mathbf{j}}_{=\mathbf{k}} + 20 \underbrace{\mathbf{i} \times \mathbf{k}}_{=-\mathbf{j}} + 20 \underbrace{\mathbf{j} \times \mathbf{k}}_{=\mathbf{i}}$$

$$= 20\mathbf{i} - 20\mathbf{j} - 10\mathbf{k}$$



$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 2 & -2 \\ 7 & 4 & 6 \end{vmatrix} = 20\mathbf{i} - 20\mathbf{j} - 10\mathbf{k}$$

5.2. Determine the distances between opposite sides of a parallelogram spanned by the vectors $\overrightarrow{AB}(6,0,1)$ and $\overrightarrow{AC}(1.5,2,1)$.

Let a be the area of the parallelogram

$$|\overrightarrow{AB} \times \overrightarrow{AC}| = a = h_1 \cdot |\overrightarrow{AB}| = h_2 \cdot |\overrightarrow{AC}|$$

$$\Rightarrow h_1 = \frac{|\overrightarrow{AB} \times \overrightarrow{AC}|}{|\overrightarrow{AB}|} = \frac{\sqrt{673}}{2\sqrt{37}}$$

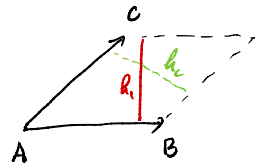
$$|\overrightarrow{AB} \times \overrightarrow{AC}| = \left| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 6 & 0 & 1 \\ 1.5 & 2 & 1 \end{vmatrix} \right| = |-2\mathbf{i} - 4.5\mathbf{j} + 12\mathbf{k}| = \sqrt{4 + \frac{81}{4} + 144}$$

$$= \frac{\sqrt{673}}{2}$$

$$|\overrightarrow{AB}| = \sqrt{37}$$

$$|\overrightarrow{AC}| = \sqrt{\frac{9}{4} + 4 + 1} = \frac{\sqrt{29}}{2}$$

$$\Rightarrow h_2 = \frac{\sqrt{673}}{\sqrt{29}}$$



5.3. Consider the vectors $\mathbf{a}(2, -3, 1)$, $\mathbf{b}(-3, 1, 2)$ and $\mathbf{c}(1, 2, 3)$. Calculate $(\mathbf{a} \times \mathbf{b}) \times \mathbf{c}$ and $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

$$\begin{aligned}
 (\mathbf{a} \times \mathbf{b}) \times \mathbf{c} &= \langle \mathbf{a} \cdot \mathbf{c} \rangle \mathbf{b} - \langle \mathbf{b} \cdot \mathbf{c} \rangle \mathbf{a} = (2 - 6 + 3) \mathbf{b} - (-3 + 2 + 6) \mathbf{a} \\
 &= - \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} - 5 \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} = \begin{bmatrix} -7 \\ 14 \\ -7 \end{bmatrix} \\
 \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) &= \langle \mathbf{a} \cdot \mathbf{c} \rangle \mathbf{b} - \langle \mathbf{a} \cdot \mathbf{b} \rangle \mathbf{c} = - \begin{bmatrix} -3 \\ 1 \\ 2 \end{bmatrix} + 7 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 10 \\ 13 \\ 19 \end{bmatrix}
 \end{aligned}$$

5.4. The points $A(1, 2, -1)$, $B(0, 1, 5)$, $C(-1, 2, 1)$ and $D(2, 1, 3)$ are given with respect to an orthonormal coordinate system. Are the four points coplanar?

$$A, B, C, D \text{ coplanar} \Leftrightarrow \underbrace{\vec{AB}}_u, \underbrace{\vec{AC}}_v, \underbrace{\vec{AD}}_w \text{ linearly dependent}$$

$\Downarrow$

$$\begin{bmatrix} -1 \\ -1 \\ 6 \end{bmatrix} \quad \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix} \quad \begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}$$

Volume of tetrahedron
 $ABCD = 0$

$$\Leftrightarrow \begin{vmatrix} -1 & -2 & 1 \\ -1 & 0 & -1 \\ 6 & 2 & 4 \end{vmatrix} = 0$$

$$12 - 2 - 8 - 2 = 0 \quad \text{true}$$

5.5. Consider the tetrahedron with vertices $A(2, -1, 1)$, $B(5, 5, 4)$, $C(3, 2, -1)$ and $D(4, 1, 3)$. Determine

a) the volume of the tetrahedron.

b) the common perpendicular of the sides AB and CD .

$$a) \quad V_d = \frac{1}{6} |\vec{AB}, \vec{AC}, \vec{AD}| = \frac{1}{6} \begin{vmatrix} 3 & 6 & 3 \\ 1 & 3 & -2 \\ 2 & 2 & 2 \end{vmatrix} = 1 - 31 = 3$$

$$b) \quad AB: \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}}_A + t \underbrace{\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}}_{\vec{AB}/3} \quad CD: \begin{bmatrix} 2 \\ 4 \\ 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 3 \\ 2 \\ -1 \end{bmatrix}}_C + \lambda \underbrace{\begin{bmatrix} 1 \\ -1 \\ 4 \end{bmatrix}}_{\vec{CD}}$$

$$\frac{\vec{AB}}{3} \times \vec{CB} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 1 & -1 & 4 \end{vmatrix} = \hat{i} \begin{vmatrix} 2 & 1 \\ -1 & 4 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 1 \\ 1 & 4 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 2 \\ 1 & -1 \end{vmatrix} = 3 \begin{bmatrix} 3 \\ -1 \\ -1 \end{bmatrix}$$

$$\text{common perp: } \begin{cases} \begin{vmatrix} x-2 & y+1 & z-1 \\ 1 & 2 & 1 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow -1(x-2) + 4(y+1) - 7(z-1) = 0 \\ \begin{vmatrix} x-3 & y-2 & z+1 \\ 1 & -1 & 4 \\ 3 & -1 & -1 \end{vmatrix} = 0 \Leftrightarrow 5(x-3) + 13(y-2) + 2(z+1) = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} -x + 4y - 7z + 13 = 0 \\ 5x + 13y + 2z - 39 = 0 \end{cases}$$

5.6. Consider the vectors $\mathbf{a}(2, 3, -1)$ and $\mathbf{b}(1, -1, 3)$. Determine the vector $\mathbf{p}$ which is orthogonal to $\mathbf{a}$ and $\mathbf{b}$ and for which $\langle \mathbf{p}, 2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k} \rangle = 51$.

Sol1

take an arbitrary vector $\mathbf{v}(x, y, z)$ and impose the

conditions: $\mathbf{v} \in \langle \mathbf{a}, \mathbf{b} \rangle^\perp \Leftrightarrow \mathbf{v} \perp \mathbf{a} \text{ and } \mathbf{v} \perp \mathbf{b}$

$$\Leftrightarrow \mathbf{v} \cdot \mathbf{a} = 0 \text{ and } \mathbf{v} \cdot \mathbf{b} = 0$$

$$\Leftrightarrow \begin{cases} 2x + 3y - z = 0 \\ x - y + 3z = 0 \end{cases}$$

this system allows us to express y and z in terms of x

$$\text{then } \mathbf{v} \in \langle \mathbf{a}, \mathbf{b} \rangle^\perp \Leftrightarrow \mathbf{v} = \mathbf{v}(x, y(x), z(x))$$

Sol2

$$\langle \mathbf{a}, \mathbf{b} \rangle^\perp = \langle \mathbf{a} \times \mathbf{b} \rangle = \left\langle \begin{bmatrix} 8 \\ -7 \\ -5 \end{bmatrix} \right\rangle = \{ (8t, -7t, -5t) : t \in \mathbb{R} \}$$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & -1 \\ 1 & -1 & 3 \end{vmatrix} = 8\hat{i} - 7\hat{j} - 5\hat{k}$$

$$p \in \langle a, b \rangle^\perp \Leftrightarrow p = t \begin{pmatrix} 8 \\ -7 \\ -5 \end{pmatrix} \text{ for some } t \in \mathbb{R}$$

$$p \cdot (2i - 3j + 4k) = 51 \Leftrightarrow (t8i - t7j - t5k) \cdot (2i - 3j + 4k) = 51$$

$$\Leftrightarrow 16t + 21t - 20t = 51$$

$$\Leftrightarrow t = \frac{51}{17} = 3$$

5.7. Consider the points $A(2, 1, -1)$, $B(3, 0, 1)$, $C(2, -1, 3)$ and $D \in Oy$. Determine the coordinates of D if the volume of a tetrahedron $ABCD$ is 5.

$$D \in Oy \Rightarrow D(0, \lambda, 0)$$

$$\text{the volume of the tetrahedron is } V = \left| \frac{1}{6} \begin{vmatrix} 1 & -1 & 2 \\ 0 & -2 & 4 \\ -2 & \lambda-1 & 1 \end{vmatrix} \right| = \left| \frac{1}{6} (-2 + 8 - 8 - 4(\lambda-1)) \right| = \left| \frac{1}{6} (2-4\lambda) \right|$$

$$\text{so } 5 = \frac{|1-2\lambda|}{3} \Leftrightarrow 15 = |1-2\lambda| \Leftrightarrow \begin{cases} \lambda = -7 \\ \lambda = 8 \end{cases}$$

5.8. Let π be a plane with normal vector $(1, 1, 1)$ intersecting the coordinate axes in A , B and C . Give formulas for the volume of the tetrahedron $OABC$ in terms of $h = d(O, \pi)$, $a = |OA|$ and $s = |AB|$ respectively.

we may assume $A(a, 0)$, i.e. $a > 0$

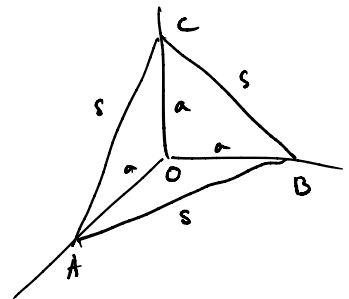
$$\pi: \frac{x}{a} + \frac{y}{a} + \frac{z}{a} = 1 \Leftrightarrow x + y + z - a = 0$$

$$h = d(O, \pi) = \frac{|a|}{\sqrt{3}} = \frac{a}{\sqrt{3}} \Rightarrow a = \sqrt{3}h$$

$$s = \sqrt{2}a \Rightarrow a = s/\sqrt{2}$$

$$V = \text{Vol}(OABC) = \frac{1}{6} \text{ volume cube with side } a$$

$$\Rightarrow V = \frac{a^3}{6} = \frac{\sqrt{3}}{2} h^3 = \frac{s^3}{12\sqrt{2}}$$



$$s/\sqrt{6}$$

$$\text{check with Heron: Area}(ABC) = \frac{\sqrt{3}}{4} s^2 \Rightarrow V = \frac{1}{3} \cdot \frac{\sqrt{3}}{4} s^2 \cdot h = \frac{s^3}{12\sqrt{2}}$$

5.9. Determine the matrices of the linear maps $\phi: \mathbb{V}^3 \rightarrow \mathbb{V}^3$ defined by $\phi(\mathbf{v}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{v}$.

$$\phi(\mathbf{i}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{i} = 2\mathbf{i} \times \mathbf{i} - \mathbf{j} \times \mathbf{i} = \mathbf{i} \times \mathbf{j} = \mathbf{k}$$

$$\phi(\mathbf{j}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{j} = 2\mathbf{i} \times \mathbf{j} = 2\mathbf{k}$$

$$\phi(\mathbf{k}) = (2\mathbf{i} - \mathbf{j}) \times \mathbf{k} = 2\mathbf{i} \times \mathbf{k} - \mathbf{j} \times \mathbf{k} = -2\mathbf{j} - \mathbf{i}$$

$$\Rightarrow [\phi] = \begin{bmatrix} \hat{\phi(\mathbf{i})}_{\downarrow} & \hat{\phi(\mathbf{j})}_{\downarrow} & \hat{\phi(\mathbf{k})}_{\downarrow} \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & -2 \\ 1 & 2 & 0 \end{bmatrix}$$

5.10. Consider the vectors $\mathbf{u} = -\mathbf{i} + 3\mathbf{j} + \mathbf{k}$ and $\mathbf{w} = \mathbf{k}$. Determine the matrix of the linear map $\phi: \mathbb{V}^3 \rightarrow \mathbb{R}$ defined by $\phi(\mathbf{v}) = [\mathbf{v}, \mathbf{u}, \mathbf{w}]$.

$$\phi(\mathbf{i}) = \begin{vmatrix} 1 & 0 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 3 \quad \phi(\mathbf{j}) = \begin{vmatrix} 0 & 1 & 0 \\ -1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1 \quad \phi(\mathbf{k}) = \begin{vmatrix} 0 & 0 & 1 \\ -1 & 3 & 1 \\ 0 & 0 & 0 \end{vmatrix} = 0$$

$$\Rightarrow [\phi] = \begin{bmatrix} \hat{\phi(\mathbf{i})}_{\downarrow} & \hat{\phi(\mathbf{j})}_{\downarrow} & \hat{\phi(\mathbf{k})}_{\downarrow} \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \end{bmatrix}$$